Oppg 1/ $X \sim N(\mu, \sigma^2)$

μ = objekt, $\sigma = 8$

$H_0: \mu = 52{,}5$ mot $H_1: \mu > 52{,}5$

$n = 25$

$\bar{x} \geq 54.0 \Rightarrow$ forkaste H_0

$$P(\text{Type I feil}) = P(\text{forkaste } H_0 \mid H_0 \text{ er sann})$$
$$= P(\bar{X} \geq 54 \mid H_0 \text{ sann})$$

$X \sim N(0,1)$

$\bar{x} \sim N(\mu, \frac{\sigma^2}{n}) \longrightarrow Z = \frac{\bar{X} - \mu}{\sqrt{\sigma^2/n}} \xrightarrow{H_0 \text{ sann}} \frac{\bar{X} - 52{,}5}{\sqrt{\sigma^2/n}}$

$$P(\bar{x} \geq 54) = 1 - P(\bar{x} \leq 54)$$
$$= 1 - P\left(\frac{\bar{x} - \mu_0}{\sqrt{\sigma^2/n}} \leq \frac{54 - \mu_0}{\sqrt{\sigma^2/n}}\right)$$
$$= 1 - P\left(Z \leq \frac{54 - 52{,}5}{8/\sqrt{25}}\right)$$
$$= 1 - \Phi(0.9375)$$
$$= 1 - 0.8212 = \underline{0{,}1788} \approx \underline{\underline{0{,}179}}$$

b) $\mu = 53{,}5$ – $P(\text{ikke forkaste } H_0 \mid H_1 \text{ er sann})$

$= P(X < 54 \mid \mu = 53{,}5)$

$= \Phi\left(\frac{54 - 53{,}5}{8/\sqrt{25}}\right) = 0.6217$

Oppg 2/ $n = 16$

$X \sim \text{binom}(n, p) = \text{binom}(16, p)$

a) hvis Jarlf ikke er synsk er sannsynligheten for å gjette riktig farge = $4/16$ $p = 1/4$

b) hvis han "rett" alle fargene bør $p > 4/16$

c) $X \sim \text{binom}(n, p)$

$P(X > c \mid p = \frac{1}{4}) \leq 0.05$

```python
from math import comb #nCr

p = 4/16
n = 16

def testobservator(x):
    s = 0
    for i in range(x,n+1):
        s += comb(n, i)*(p**i)*(1-p)**(n-i)

    return s

#Oppgave c)
c = 0 #startverdi
while testobservator(c) > 0.05:
 c += 1

c -= 1 #hvis f.eks. 3 er den siste verdien som oppfyller ulikhet,
#vil while-løkken kjøre, og dermed gi en for høy verdi
print(f"Den kritiske verdien som gjør at signifikansnivået er 0.05 er: {c}")
```

```
Den kritiske verdien som gjør at signifikansnivået er 0.05 er: 7
```

e) $x_{obs} = 5$

$p = P(X \geq 5 \mid H_0 = \text{sann})$

$p = 1 - P(X < 4 \mid p = 1/4)$ ← blå tabell under binom, $n = 16$, $x = 5$, $P = 0.25$

$= 1 - 0.630$

$p = 0.370$

f) $P(\text{forkaste } H_0 \mid H_0 = \text{riktig}) \leq \alpha$

$= P(X > 7 \mid p = 0.25) \leq \alpha$

```python
y=0
while testobservator(y) > 0.05:
 y += 1
print(f"Verdi som oppfyller likningen {testobservator(y)}")
```

$\Rightarrow 0.027$

Oppg 3/

```python
#STACK_10_Oppg3.py
import numpy as np
my_0 = 16

x = np.array([15.61, 15.65, 16.24, 16.33, 16.45, 16.32])

snitt = np.mean(x)
n = len(x)

s2 = 1/(n-1)*sum([(i-snitt)**2 for i in x])
s = np.sqrt(s2)

def testobservator():
    global teller, nevner
    teller = snitt - my_0
    nevner = s/np.sqrt(6)

    return teller/nevner



#Oppgave a)
#P(T>k) = 0.05
#P(T=<k) = 0.95
#Finner kritisk verdi for t-fordeling når v = n-1 og alfa = 0.0
#k = 2.015

#Oppgave b)
#print(testobservator()) => 0.661...

```

c) $T_{obs} = 0.6613 = \dfrac{\bar{x} - \mu_{obs}}{s/\sqrt{6}}$

$\Rightarrow \mu_{obs} = \bar{x} - 0,6613 \cdot \dfrac{s}{\sqrt{6}} \Rightarrow \mu_{obs} = 16,000\ldots$

$H_0 = \text{sann} \Rightarrow$ Ikke forkast

Oppg 4/

$y \sim N(\mu, \sigma^2)$

Påstand: $\sigma < 2$

a) $\Rightarrow H_0: \sigma \leq 2 \quad = \quad \sigma = 2$

b) $H_1: \sigma > 2$

d) $n = 10 \quad , \quad s = 3,02$

$$\frac{(n-1)s^2}{2^2} = \text{testobservator} : \sim \chi^2_{n-1}$$

$$\frac{9 \cdot 3,02^2}{4} = 20,52$$

e) $T = \frac{(n-1)s^2}{2^2} \sim \chi^2_{n-1}$

$P(T > k) = 0,05$

$\Rightarrow P(T < k) = 0,95 \rightarrow \begin{matrix} \alpha = 0,05 \\ \nu = n-1 = 9 \end{matrix} \} 16,919$

f) $T = 16,919 = \frac{9 \cdot s^2}{\sigma^2} \Rightarrow \sigma = \sqrt{\frac{9 \cdot s^2}{16,919}}$

$\sigma = \frac{3 \cdot 3,02}{\sqrt{16,919}} = 2,203$ | $H_1 = \text{scan}$

$= \underline{\text{forkast } H_0}$

g) $y = 64°C$

$y \sim N(\mu, \sigma^2)$, begge ukjent

$T = \frac{\bar{x} - \mu}{\sqrt{s^2/n}}$, $P\left(-t_{\frac{\alpha}{2}} < \mu < t_{\frac{\alpha}{2}}\right) = 1 - \alpha$

$\Rightarrow P\left(\bar{x} - t_{\frac{\alpha}{2}} s \cdot \frac{1}{\sqrt{n}} < \mu < \bar{x} - t_{\frac{\alpha}{2}} s \cdot \frac{1}{\sqrt{n}}\right)$

$\left|t_{\frac{\alpha}{2}}\right|_{\substack{\alpha = 0,05 \\ \nu = 10-1}} = 2,262$, $s = 3,02$, $\bar{x} = 64$, $n = 1$

$\Rightarrow$ Konfidensintervall: $\underline{\left[57.169, 70,831\right]}$